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#include "UCNEC.h"
int32_t temp = 0;
UCNEC_myIR(A0);
#define F 16718055
#define B 16732845
#define L 16713975
#define R 16734885
#define S 16719075

// Motor control pins: L298N H bridge
const int enAPin = 6; // Left motor PWM speed control
const int in1Pin = 7; // Left motor Direction 1
const int in2Pin = 5; // Left motor Direction 2
const int in3Pin = 4; // Right motor Direction 1
const int in4Pin = 2; // Right motor Direction 2
const int enBPin = 3; // Right motor PWM speed control

int command; //bien luu trang thai Bluetooth
int speedCar = 160; // toc do 50 - 255.
int speed_Coeff = 2; // he so suy giam <=> giam may lan toc do

/*****Define Line Track
pins*****/
const int SensorLeft = 10; //Left sensor input (A5)
const int SensorMiddle = 9; //Midd sensor input (A4)
const int SensorRight = 8; //Right sensor input (A3)
int SL; //Status of Left line track sensor
int SM; //Status of Midd line track sensor
int SR; //Status of Righ line track sensor
void goAhead(int speedcar1, int speedcar2){

    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,HIGH); // LEFT tien
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,HIGH); // RIGHT tien
    analogWrite(enAPin, speedcar1);
    analogWrite(enBPin, speedcar2);

}

void goBack(int speedcar1, int speedcar2){
    digitalWrite(in1Pin,HIGH);
    digitalWrite(in2Pin,LOW); // LEFT lui
    digitalWrite(in3Pin,HIGH);
    digitalWrite(in4Pin,LOW); // RIGHT lui
    analogWrite(enAPin, speedcar1);
    analogWrite(enBPin, speedcar2);
}

void goRight(int speedcar1, int speedcar2){
    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,HIGH); // LEFT tien
    digitalWrite(in3Pin,HIGH);
    digitalWrite(in4Pin,LOW); // RIGHT lui

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        analogWrite(enAPin, speedcar1);
        analogWrite(enBPin, speedcar2);

    }

void goLeft(int speedcar1, int speedcar2){
    digitalWrite(in1Pin,HIGH);
    digitalWrite(in2Pin,LOW);          // LEFT lui
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,HIGH);        // RIGHT tien
    analogWrite(enAPin, speedcar1);
    analogWrite(enBPin, speedcar2);

}

void stopRobot(){
    digitalWrite(in1Pin,LOW);
    digitalWrite(in2Pin,LOW);
    digitalWrite(in3Pin,LOW);
    digitalWrite(in4Pin,LOW);
    analogWrite(enAPin, 0);
    analogWrite(enBPin, 0);
}

void setup() {

    pinMode(enAPin, OUTPUT);
    pinMode(in1Pin, OUTPUT);
    pinMode(in2Pin, OUTPUT);
    pinMode(in3Pin, OUTPUT);
    pinMode(in4Pin, OUTPUT);
    pinMode(enBPin, OUTPUT);

    Serial.begin(9600); //Init Serial port rate set to 9600

    pinMode(SensorLeft, INPUT);    //Init left sensor
    pinMode(SensorMiddle, INPUT);  //Init Middle sensor
    pinMode(SensorRight, INPUT);   //Init Right sensor
}

void loop() {
    SL = digitalRead(SensorLeft);
    SM = digitalRead(SensorMiddle);
    SR = digitalRead(SensorRight);

    //right fast 20 200
    if (SM == HIGH && SR == HIGH && SL == LOW)
    {
        goAhead(160,40);delay(100);
    }
    if (SM == HIGH && SR == LOW && SL == HIGH)
    {
        goAhead(40,160);delay(100);
    }
}

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    if (SM == HIGH && SR == LOW && SL == LOW)
    {
        goAhead(160,160);
    }
    if(SM == LOW && SR == LOW && SL == HIGH)
    {
        goAhead(80,200);delay(100);
    }
    if(SM == LOW && SR == HIGH && SL == LOW)
    {
        goAhead(200,80);delay(100);
    }
    else
    {
        stopRobot(); delay(100);
    }
}
```